

CogniBoost — Master Plan

Question 4 Response: Engineer Headcount & Timeline

From CogniBoost engineering team to Coral Lozano · May 13, 2026 · cogniboost.co

You asked for three numbers before deciding on engineer headcount: (1) time to complete the full v2.0 scope with **one engineer + aggressive Claude Code productivity**, (2) time with **two engineers + aggressive Claude Code productivity**, and (3) approximate **monthly cost** for each option, with a target to launch **before September**.

Status update: Phase 0 (the one-week foundation week) is already shipped. The Claude API integration, the writing-grader module with the full CEFR rubric A1–C2, the v2.0 schema additions (*submissions, vocabulary, vocabulary_mastery, lab_topics, lab_sessions, lab_registrations, lab_feedback, listening_assessments, reading_passages, certificates*), and the smoke test are all in production on cogniboost.co as of May 13, 2026 (PR #7). Below is the full remaining plan and the two paths to ship it.

Remaining Scope (post-Phase 0)

The work that still needs to ship to call v2.0 launched. All sized as net-new engineering hours; Phase 0 is excluded.

Phase	Deliverables	Standard est.	With Claude Code
1	Submission API · writing editor · teacher dashboard skeleton · Lesson Library · Lab sessions + feedback form	~8 weeks	~5–6 weeks
2	Reading + Listening graders · Vocab mastery (SRS) · Student progress dashboard · Annotation tools · One-Click Lesson + Lab Pack generators	~8 weeks	~5–6 weeks
3	Certification system + public verify · cohort/level analytics · automated nudges · PWA polish · Anthropic billing transition	~6 weeks	~4 weeks
Total	Complete v2.0 platform	~22 weeks	~14–16 weeks

The compression with aggressive Claude Code productivity is 30–35% — driven by schema scaffolding, boilerplate API endpoints, React-component scaffolding via shadcn/ui, and prompt-engineering iteration. Testing and integration time compress less.

Option A — One Engineer + Aggressive Claude Code

Phase	Duration	Cumulative	Calendar (from May 13)
1	~5–6 weeks	Weeks 1–6	May 13 → mid-June
2	~5–6 weeks	Weeks 7–12	mid-June → late July
3	~4 weeks	Weeks 13–16	late July → late August
Total	~14–16 weeks	~3.5–4 months	Launch ~late August

Verdict — barely makes September. If anything slips by even a week (scope creep on Lesson Library, AI-grader prompt tuning takes longer to settle, a third-party dependency surprise), launch moves into September. Doable, but tight. Best for a launch-readiness target of **September 1**.

Option B — Two Engineers + Aggressive Claude Code

Two engineers in parallel. One owns backend + schema + AI grading; the other owns frontend + dashboards + teacher portal. Coordination overhead at ~15% (PR reviews, schema alignment), so the speedup is not 2×.

Phase	Duration	Cumulative	Calendar (from May 13)
1	~3 weeks	Weeks 1–3	May 13 → early June
2	~3 weeks	Weeks 4–6	early June → late June
3	~2.5 weeks	Weeks 7–9	late June → mid-July
Total	~8–9 weeks	~2 months	Launch ~mid-July

Verdict — comfortably beats September. Mid-July launch leaves ~6 weeks of buffer for soft-launch, Coral content production, and customer onboarding before the September window. Recommended if budget permits and the September deadline matters.

Approximate Monthly Cost

Cost depends on the engagement model. The numbers below use a placeholder hourly rate so you can pick the structure that works for you; final pricing is at your discretion and is not yet committed.

Engineering effort + Anthropic API + infra

Item	Option A (1 eng)	Option B (2 eng)
Engineering hours / month	~160 hrs/mo	~320 hrs/mo (split across 2)
Engineering cost / month [TO BE FILLED]	\$X / month	\$2X / month
Anthropic Claude API (Sonnet + Opus)	\$30–60	\$30–60
Existing infra (Railway, Vercel, GCS, Stripe fees, Resend, Sentry)	~\$50	~\$50
Total monthly burn	\$X + ~\$100	\$2X + ~\$100

Total project cost (engineering only)

Option	Duration	Eng hours total	Eng cost total [TO BE FILLED]
A — 1 engineer	~3.5–4 months	~600 hrs	\$X × 600 hrs
B — 2 engineers	~2 months	~640 hrs	\$X × 640 hrs

Note: total engineering hours are roughly comparable across both options — two engineers in parallel save calendar time, not total hours. The choice is between *longer engagement at lower monthly burn* (Option A) vs *shorter engagement at higher monthly burn but earlier launch* (Option B). Anthropic API usage is the same in both.

Recommendation

Option B — Two engineers, if the September launch target matters. The ~\$X/month additional monthly cost buys ~6 weeks of de-risking buffer before the deadline, plus parallel work means a frontend engineer can polish UX while the backend engineer optimizes grading throughput.

Option A — One engineer is acceptable if the September date is soft and lower monthly burn is the priority. The August finish is real but fragile — any one-week slip pushes into September.

Decision Request

Pick one. We can kick off Phase 1 immediately on confirmation.

Choice
☐ Option A — One engineer + aggressive Claude Code (launch ~late August, fragile)
☐ Option B — Two engineers + aggressive Claude Code (launch ~mid-July, recommended)
☐ Other (free text — please describe)

Next Step

1. You confirm Option A or B and we lock the kickoff cadence.
2. First Phase 1 deliverable is the Submission API + writing editor — end-to-end visible on cogniboost.co within 5 working days.
3. If you want a fixed-fee structure rather than hourly, we can quote that off this scope. Range of acceptable risk on a fixed fee is wider — we'd ask for a +20% buffer to absorb the inherent uncertainty in v2.0 features that haven't been built before (Lab Packs, vocabulary-mastery SRS, Spanish L1 grading patterns).

This document accompanies the revised *MASTER_PLAN_REVISED.md* in the project repo (reflecting Coral's locked decisions Q1, Q2, Q3, Q5, Q6, Q7 + v2.0 academic-model updates). Question 4 is the last gating decision before Phase 1 kickoff.